

IMPLEMENTATION OF A WEB-BASED SHS RECORD MANAGEMENT SYSTEM

A Capstone Project Proposal Presented to the
Faculty of the College of Computer Studies
Benedicto College - Mandaue Campus
A.S Fortuna St. Mandaue City, Cebu

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology

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July 2026

Chapter 1

THE PROBLEM AND ITS SCOPE

Rationale

The integration of information technology in educational institutions has significantly improved how academic records and student services are managed. However, at Cebu Far East Institute (CFEI), Senior High School students are still required to personally visit designated school offices to access important academic information such as grades, study loads, tuition balances, and to request official documents. This process results in repeated manual inquiries, increased workload for school personnel, and delays in accessing student-related information.

Teachers also face challenges in managing and submitting student grades efficiently, as the current process lacks a structured digital workflow. Administrators, on the other hand, must handle student records, document requests, enrollment, and announcements manually through fragmented processes.

To address these concerns, this study proposes the development of INFORM — a Web-Based Student Academic Records and Service Management System for Senior High School at CFEI. The system provides a centralized and secure online platform where students can access their grades, study loads, class schedules, tuition balances, and request official documents anytime through an internet connection. It also provides teachers with a structured grade management workflow and gives administrators centralized control over student records, enrollment, document processing, and announcements.

Theoretical Background

The Information System Theory (Leavitt & Whisler, 1958) explains how the integration of people, technology, and processes facilitates the collection, processing, storage, and dissemination of information within an organization. According to Leavitt and Whisler (1958), information systems play a vital role in decision-making by converting raw data into meaningful information that supports management and operational activities. This theory emphasizes the importance of designing systems that ensure data accuracy, timeliness, and accessibility, allowing institutions to achieve efficiency and effectiveness in their daily operations.

In relation to the proposed Web-Based Student Academic Records and Service Management System, this theory highlights how a well-structured information system can streamline academic record and student service management. By providing a centralized platform, the system allows students to access grades, study loads, balances, announcements, and other student-related information efficiently. This not only enhances data reliability but also simplifies administrative processes, promoting transparency and efficiency between students and school personnel.

Systems Theory (Ludwig von Bertalanffy, 1968). Systems Theory views an organization as a collection of interrelated and interdependent parts working together to achieve a common goal. According to Bertalanffy (1968), the success of a system depends on the harmony and functionality of its subsystems. When applied to digital environments, this theory emphasizes that every component whether human, software, or hardware must work cohesively to ensure smooth operation and optimal results.

The proposed Web-Based Student Academic Records and Service Management System embodies this theory through its interconnected modules such as login authentication, academic record access, announcement management, and dashboard navigation. Each module functions as a subsystem that contributes to the overall performance of the system. When these components interact effectively, the system ensures that information flows seamlessly, improving user experience and operational efficiency.

Human - Computer Interaction (HCI) Theory (Card, Moran, & Newell, 1983) examines how people interact with computer systems and how design elements can improve usability, accessibility, and satisfaction. According to Card, Moran, and Newell (1983), an effective system must balance functionality and user experience to ensure that users can complete their tasks easily and intuitively. HCI theory also highlights the importance of feedback, simplicity, and clarity in system interfaces.

Applied to the proposed Web-Based Student Academic Records and Service Management System, this theory ensures that the system is user-friendly, accessible, and efficient for students, teachers, and administrators. The system presents information in a structured and understandable web-based interface, allowing users to navigate features and access services conveniently. By prioritizing usability, clarity, and accessibility, the system minimizes user confusion and improves overall user experience in accessing academic records and student-related services.

Management Information System (MIS) Theory (Davis & Olson, 1985) focuses on how computer-based systems support organizational management by

providing accurate, relevant, and timely information for planning and decision-making. Davis and Olson (1985) assert that MIS helps organizations achieve efficiency by automating information flow and minimizing redundant manual processes. The theory also underscores the role of technology in improving productivity and decision accuracy.

For the proposed Web-Based Student Academic Records and Service Management System, the MIS Theory is relevant because the system functions as an academic management and student service tool. It automates the retrieval and management of student information, allowing students, teachers, and administrators to access accurate data quickly and efficiently. This automation minimizes clerical errors and enhances the institution's ability to maintain consistent and updated records, aligning with the principles of effective information management.

Data Security Theory (Baskerville, 1993) emphasizes the protection of information from unauthorized access, modification, or destruction. Baskerville (1993) explains that information systems must uphold three key security principles: confidentiality, integrity, and availability. These ensure that sensitive information remains private, unaltered, and accessible only to authorized users. The theory highlights that security mechanisms must be proactive to prevent data breaches and maintain trust between users and system administrators.

In the proposed Web-Based Student Academic Records and Service Management System, this theory is reflected in the system's security features such as secure login authentication, password protection, restricted access, and centralized database security. These measures protect student information from unauthorized access while maintaining user confidence in the system. By prioritizing

data protection and privacy, the system fosters a secure environment that aligns with institutional goals for security, accountability, and efficient information management.

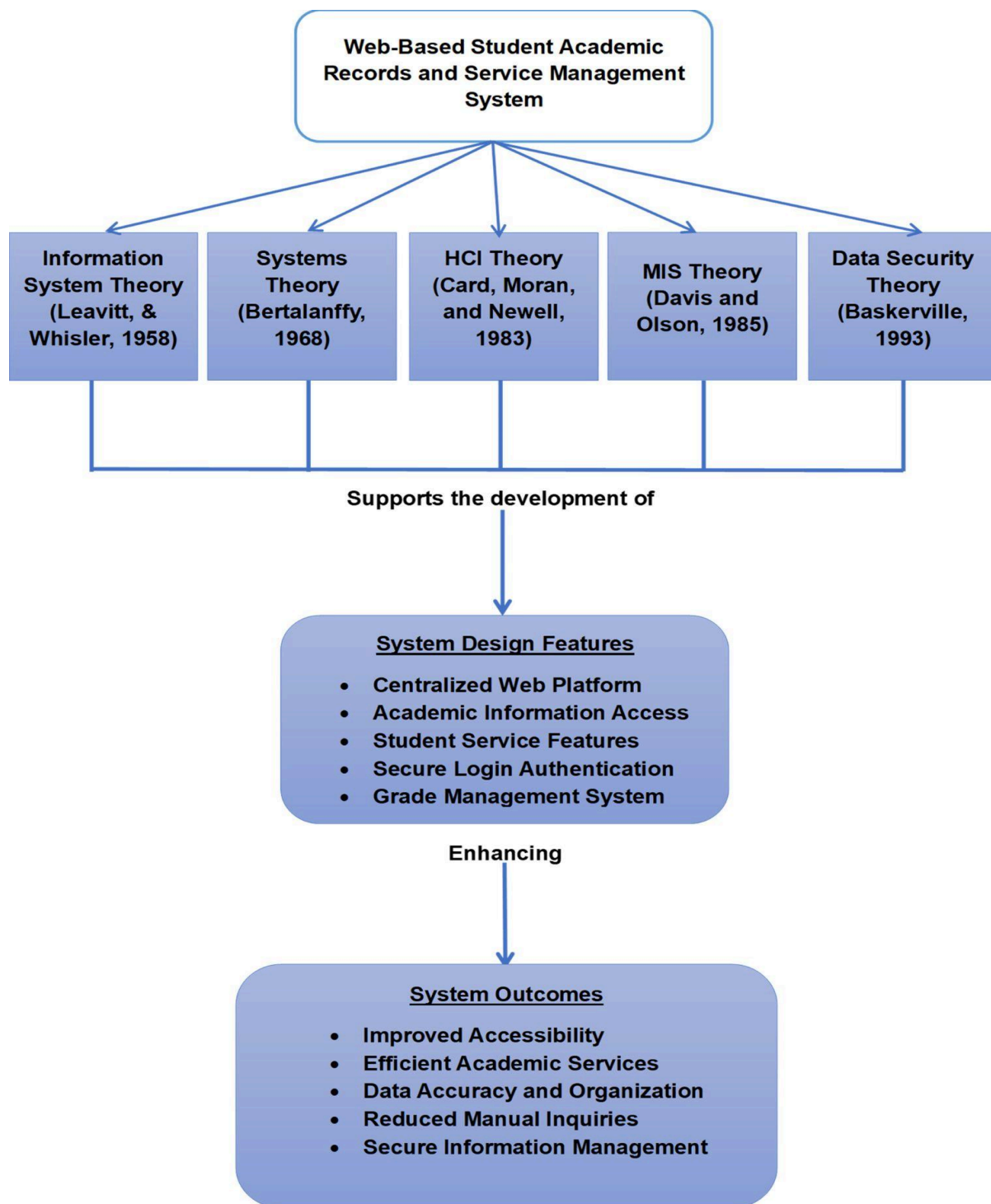


Figure 1

Theoretical Framework

Statement of the Problem

Senior High School students at Cebu Far East Institute (CFEI) lack a centralized online platform for accessing academic records and student-related services, resulting in repeated manual inquiries, delays in retrieving information, and increased workload for school personnel.

Specifically, this study seeks to answer the following questions:

1. What are the current challenges faced by SHS students, teachers, and administrators in accessing and managing academic records and student services at CFEI?
2. What features and functionalities are necessary in a web-based system to effectively centralize the management of student academic records and services for Senior High School?
3. How will the proposed system improve the accessibility and efficiency of managing student academic records and services compared to the current manual process?
4. To what extent does the proposed system meet the needs of students, teachers, and administrators in terms of usability, security, and functionality?

General Problem of the Study

The absence of a centralized and accessible online platform for managing student academic records and services at Cebu Far East Institute (CFEI) results in repeated manual inquiries, delays in retrieving academic information, and increased workload for both students and school personnel.

Specific Problem of the Study

1. What are the challenges faced by Senior High School (SHS) students, teachers, and administrators in accessing and managing academic records and student-related services at Cebu Far East Institute (CFEI)?
2. What features are needed in a web-based system to centralize and improve the management of student academic records and the delivery of student-related services?
3. How will the proposed system improve the accessibility and efficiency of managing student academic records and services compared to the current manual process?
4. To what extent does the proposed system meet the needs of students, teachers, and administrators in terms of usability and functionality?

General Objective of the Study

The study aims to develop and implement INFORM, a Web-Based Student Academic Records and Service Management System that provides Senior High School (SHS) students, teachers, and administrators of Cebu Far East Institute (CFEI) with a centralized, secure, and accessible online platform for managing academic records and student-related services.

Specific Objectives of the Study

Specifically, this study aims to:

1. Identify the current challenges in accessing and managing student academic records and services among Senior High School (SHS) students, teachers, and administrators at Cebu Far East Institute (CFEI).
2. Design and develop a web-based system that allows SHS students to securely access their grades, class schedules, study loads, tuition balances, and announcements through an online portal.
3. Implement a structured grade management workflow that enables teachers to accept grade requests, compute grades, and submit them for administrative verification before releasing them to students.
4. Provide administrators with centralized tools for managing student enrollment, grade verification, document request processing, tuition records, and school announcements.
5. Evaluate the effectiveness of the proposed system in improving the accessibility, accuracy, and efficiency of student academic record management and service delivery at Cebu Far East Institute (CFEI).

Scope and Limitation of the Study

Scope

The study focuses on the design and development of INFORM, a Web-Based Student Academic Records and Service Management System for Senior High School students and personnel at Cebu Far East Institute (CFEI).

The system serves three primary user groups: students, teachers, and administrators. Students can securely log in to access their grades, class schedules, study loads, tuition balances, announcements, and submit requests for official

documents such as the Transcript of Records, Certificate of Enrollment, and Good Standing Certificate. The system also includes JOBERT, an AI-powered chat assistant that responds to student inquiries regarding grades, tuition, and enrollment through the online portal.

Teachers can manage and process student grade requests through a structured five-step workflow: student request, teacher grade computation, submission to the administrator, administrative verification, and final release of grades to the student. Teachers can also monitor student attendance and manage document endorsements.

Administrators have centralized access to student and teacher records, enrollment management, grade request verification, document request processing with release date assignment, tuition monitoring, and school-wide announcement management. The system also includes a teacher time log feature that enables administrators to monitor teacher attendance on campus in real time.

The system incorporates data privacy measures in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). Access to student records is restricted based on user roles through secure login authentication, and all user passwords are encrypted to protect sensitive academic information from unauthorized access.

Limitation

The proposed system is limited to the management and display of student academic records and selected student-related services for Senior High School at

Cebu Far East Institute (CFEI). It does not process online payments or financial transactions; it only displays tuition balances and payment status.

The system does not perform automatic grade computation. Teachers are responsible for manually computing final grades before submitting them through the system. The grade management module facilitates the workflow and approval process rather than performing the grade calculations.

The system requires a stable internet connection and accurate data entry from authorized users. Any interruption in internet connectivity or incorrect data input may affect the reliability and accuracy of the system. Access is strictly limited to registered users—students, teachers, and administrators—with valid system accounts.

Significance of the Study

The Web-Based Student Academic Records and Service Management System provides several benefits to different stakeholders within educational institutions.

For **School Administration**, the proposed system helps improve the organization, accessibility, and management of student records through a centralized database system. It reduces administrative workload, improves record accessibility, and minimizes the risks of data loss, duplication, and manual processing errors.

For **Educational Institutions**, the implementation of the proposed system supports the modernization of academic services through the integration of web-based technologies in managing student information and student-related services. It

promotes efficiency, accessibility, organization, and improved delivery of academic services.

For **Teachers**, the system allows authorized teachers to input and manage student grades through a centralized platform. This improves the efficiency and accuracy of recording student performance while minimizing paperwork and delays in grade processing.

For **Students**, the system provides a convenient and accessible platform for viewing academic information such as grades, enrolled subjects, study loads, tuition balances, announcements, and other relevant records online. This allows students to monitor their academic progress more efficiently while reducing the need for repetitive manual inquiries and office visits.

For **Future Researchers**, this study may serve as a reference for future studies related to web-based student information systems, academic record management, and student service platforms. The study may also provide insights into how web-based technologies can improve accessibility, organization, and management of student-related information and services.

Chapter 2

REVIEW OF RELATED LITERATURE

Related Literature

“Designing and Implementing E-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines” by Grepon et al. (2021) The study discussed the development of an e-school system designed to address the lack of computerized platforms for managing student records and school-related services. The researchers found that the system improved the efficiency of school transactions and centralized the management of academic information. The study also emphasized the importance of web-based platforms in improving accessibility and communication among students, teachers, and administrators. This literature supports the present study because it demonstrates how digital systems improve the accessibility, organization, and management of student academic information and services within educational institutions.

In the article **“Student Registration and Records Management Services Towards Digitization” by Falolo, Capillas, Vergarra, and Cerbito (2022)** The significance of digitizing student record management systems in educational institutions was discussed. The authors emphasized that delays in retrieving academic records often occur when paper-based record management systems are utilized. The article also highlighted the possibility of data inaccuracies and inefficiency when manual systems are used. Furthermore, the study emphasized the importance of implementing digital and web-based systems to improve accessibility, organization, and management of academic information. This literature is relevant to

the proposed Web-Based Student Academic Records and Service Management System because it highlights the need for centralized digital platforms that improve accessibility, accuracy, and efficiency in managing student academic records and services.

Daim, Gungor, Basoglu, Yarga, and VanDerSchaaf (2024), in their research entitled **“Exploring Student Information Management System Adoption Post Pandemic: Case of Turkish Higher Education,”** investigated the role of higher educational institutions in managing student information using student information management systems. The research concluded that such systems help store and process information regarding student details, course schedules, grades, and financial information while reducing administrative workload. The study also highlighted the significance of digital systems in improving accessibility, accuracy, and efficiency of information management. This literature is relevant to the proposed system because it emphasizes the importance of web-based student information systems in improving the management and accessibility of academic records and student-related services.

In the article **“Management of Educational Institutions through Information Systems for Enhanced Efficiency and Decision-Making,”** Sunarjo, Chakim, Maulana, and Fitriani (2024) discussed the importance of information systems in improving the efficiency of educational institutions. The authors explained that digital information systems facilitate the effective management of academic information while enhancing communication among students, teachers, and administrators. The article concluded that implementing information systems improves efficiency, accessibility, and decision-making processes within educational institutions. This literature is relevant to the proposed Web-Based Student Academic

Records and Service Management System because it highlights the advantages of implementing centralized and web-based information systems for managing academic information and student-related services.

In the article **“The Role of Centralized Databases in Modernizing Campus Infrastructure,” Balcita and Palaoag (2020)** highlighted the profound correlation between technological facility upgrades and student success rates. The researchers emphasized that deploying an integrated school management network driven by a consolidated data architecture is essential for elevating the baseline quality of institutional operations. The study concluded that modernizing campus networks optimizes service delivery while systematically eliminating structural administrative bottlenecks. This literature is directly relevant to the proposed system because it underscores the necessity of moving away from manual setups toward unified data frameworks to maximize service efficiency.

Related Studies

The study **“Design and Implementation of a Student Information System by Ike (2023),”** developed a computerized system for managing student academic records through a centralized database. The results showed that implementing the system improved the accuracy of student information and reduced the time required to retrieve academic records. The study concluded that automated systems improve administrative efficiency in educational institutions. This study supports the proposed Web-Based Student Academic Records and Service Management System because it highlights the importance of centralized and automated systems in improving accessibility, organization, and management of student-related information and services.

The study “**IntelliSchool: A Student Information System for Senior High School by Campanan et al. (2024)**,” developed a web-based platform for managing student academic records and grading information. The system allowed teachers to input grades and enabled students to access their academic records online. The findings showed that the system improved transparency, accessibility, and efficiency in managing student information. This study supports the development of the proposed system because it demonstrates the effectiveness of web-based student information systems in providing centralized and accessible academic services.

The study “**Design and Testing of a Web-Based Student Information Management System by Pasaribu and Argadikusuma (2024)**,” developed a digital platform that allows institutions to manage student academic records through an online database. The researchers found that the system improved the efficiency of managing student data and allowed faster retrieval of academic information. The study also highlighted the importance of web-based technologies in improving accessibility and management of academic services. This study supports the present research because it demonstrates the advantages of implementing web-based systems for managing student academic records and services.

The study “**Development of a Student Records Management System” by Duruin (2024)** focused on designing and implementing a computerized platform for managing student academic records. The researcher applied the Software Development Life Cycle (SDLC) in developing the system. The results showed that the developed system improved the efficiency of storing, retrieving, and managing academic records compared to manual record management. The study also highlighted the effectiveness of digital systems in improving accessibility and organization of student information. This study supports the present research

because it demonstrates how web-based and centralized systems improve the management of student academic records and services. .

Another local study entitled **“Development of a Streamlined Student Information and Grading System for Surigao del Norte College of Agriculture and Technology” by Orboc (2025)**. The researcher developed a system that integrates student information management and grading processes into one platform. The system allowed teachers to input grades while enabling students to access their academic records online. The results showed that the system improved the efficiency of academic record management and reduced administrative workload. This study is related to the proposed research because it demonstrates the effectiveness of centralized and web-based student information systems in improving accessibility and management of student academic services.

Chapter 3

TECHNICAL BACKGROUND

The development of the proposed Web-Based Student Academic Records and Service Management System requires the use of appropriate frontend and backend technologies to ensure efficiency, reliability, accessibility, and maintainability. The selected technologies will support the creation of a responsive and user-friendly interface, efficient server-side processing, secure handling of academic records, and effective communication between the system components. Through the integration of these technologies, the proposed system is expected to provide a more organized, accessible, and efficient platform for managing student academic information and related services.

Frontend Technologies

Next.js

Next.js is a React-based web development framework used for building modern, scalable, and high-performance web applications. It provides features such as server-side rendering, routing, and optimized performance that improve the speed, efficiency, and responsiveness of web systems.

In the proposed Web-Based Student Academic Records and Service Management System, Next.js will be used to develop the frontend interface for students, teachers, and administrators. It enables the development of organized, maintainable, and interactive web pages that support efficient access to academic records and student-related services.

HTML

HTML (HyperText Markup Language) is the standard markup language used for creating the structure and content of web pages. It provides the foundation of web applications by organizing elements such as text, forms, buttons, tables, and navigation components.

In the proposed Web-Based Student Academic Records and Service Management System, HTML will be used to structure the web pages and organize the presentation of academic records, student services, announcements, and other system functionalities.

Bootstrap

Bootstrap is a widely used frontend framework that provides pre-designed user interface components and responsive layout utilities for web development. It enables developers to efficiently create responsive and visually organized interfaces without extensive custom styling.

The framework includes ready-made components such as forms, navigation bars, buttons, dashboards, and tables that improve usability and interface consistency. In the proposed system, Bootstrap will be used to create a responsive and user-friendly interface for accessing academic records and student-related services across different devices and screen sizes.

Typescript

TypeScript is a programming language developed by Microsoft that extends JavaScript by adding static typing and advanced development features. It helps

developers create cleaner, more maintainable, and scalable applications while reducing coding errors during development.

In the proposed system, TypeScript will be utilized to improve code organization, maintainability, and reliability in developing frontend functionalities and interactive features within the web application.

Backend Technologies

Node js

Node js is an open-source JavaScript runtime environment built on Chrome's V8 engine that allows developers to run JavaScript code outside the browser. It is designed for building fast, scalable, and efficient server-side applications, especially for web systems and real-time applications.

In the proposed system, Node.js will be utilized as the backend runtime environment to handle server-side operations, manage API requests, process data, and support communication between the frontend application and the database efficiently.

Express js

Express.js is an open-source and lightweight web application framework for Node.js that simplifies the development of server-side applications and APIs. It provides tools and features for handling routing, requests, responses, middleware, and server management, making web application development faster, more organized, and efficient.

In the proposed system, Express.js will be utilized to manage backend functionalities such as handling API routes, processing client requests, managing server

responses, and facilitating communication between the frontend application and the database efficiently.

Sequelize

Sequelize is an open-source Object-Relational Mapping (ORM) library for Node.js that simplifies database management and interaction using JavaScript or TypeScript. It allows developers to perform database operations such as creating, reading, updating, and deleting data without writing complex SQL queries directly, making development more efficient and organized.

In the proposed system, Sequelize will be utilized to manage database operations, define data models and relationships, handle queries efficiently, and facilitate secure communication between the backend application and the database.

API Development and Communication

Rest API

A Representational State Transfer Application Programming Interface (REST API) is an architectural style that defines a set of constraints for creating stateless web services. It serves as the vital communication bridge in full-stack applications, allowing the frontend user interface to interact seamlessly with the backend server and database. Built entirely on top of the standard HTTP protocol, a REST API uses uniform resource identifiers (URIs) to locate specific resources and relies on standard HTTP methods specifically GET to read, POST to create, PUT to update, and DELETE to remove data to execute fundamental CRUD operations.

Postman

Postman is an industry-standard, collaborative API (Application Programming Interface) platform designed to simplify the development, testing, and documentation of web services. In modern full-stack development, backends and frontends interact via APIs. Postman acts as an intermediary client tool that allows developers to construct, send, and analyze HTTP/HTTPS requests to an API backend without needing a completed frontend user interface.

Development tools and Database Management

Mysql

Mysql is an open-source Relational Database Management System (RDBMS) developed and maintained by Oracle Corporation. It operates on a client-server architecture where the database server handles data storage, querying, and manipulation, while client applications communicate with the server via structured requests. Mysql is widely recognized for its speed, reliability, and high scalability, making it a staple backend component in modern full-stack web applications.

Hardware Requirements

- Laptop: HP Elitebook 835 G7 Notebook PC - a compact business laptop powered by an AMD Ryzen processor, designed for fast performance, portability, and productivity.
- Ram: 16 GB DDR4 3200MHZ - Provides smooth multitasking and fast data processing for development workflows.
- Storage: 512 SSD - offers high-speed storage for quick application loading, build process, and data access.

- Operating System: Windows 11 Pro 64-bit - Your computer's operating system that manages software, security, and multitasking.
- Graphics: AMD Radeon Graphics - The graphics engine that handles visuals, gaming, videos, and GPU-related tasks.

Chapter 4

Methodology

Project Feasibility

The feasibility of INFORM was assessed across five dimensions to determine whether the project is realistic, sustainable, and beneficial to the Cebu Far East Institute (CFEI) prior to full implementation.

Technical Feasibility

The system was built using free and open-source technologies, including Next.js, React, TypeScript, Node.js, Express.js, Sequelize, and MySQL, all of which are well-documented, widely used, and within the capability of the development team. The system follows a client-server architecture with a RESTful API that connects the frontend and backend components. User authentication is implemented using JSON Web Tokens (JWT) and bcrypt.js password encryption, which are recognized industry standards for web application security. The hardware used during development a laptop equipped with 16 GB of RAM, a 512 GB SSD, and Windows 11 Pro is sufficient for developing, running, and testing the system. Deployment requires only a basic web server capable of hosting a Node.js application with a MySQL database, making it suitable for implementation within the school's existing computing environment. Therefore, the proposed system is considered technically feasible.

Operational Feasibility

INFORM is designed to replace the manual and office-based processes that are currently practiced at Cebu Far East Institute (CFEI). The system serves three

primary user roles students, teachers, and administrators each with a dedicated portal that reflects the school's existing workflows. Students can access their grades, class schedules, study loads, tuition balances, official document requests, and school announcements without the need to visit school offices. Teachers can manage grade requests through a structured five-step workflow consisting of student request, teacher grade computation, submission to the administrator, administrative verification, and final release of grades to the student. Administrators can centrally manage student enrollment, grades, document requests, tuition records, and school announcements while maintaining full audit control over administrative actions. The system features a responsive user interface that is accessible on both desktop and mobile devices and includes light and dark mode options to enhance user comfort and accessibility. Therefore, the proposed system is considered operationally feasible.

Economic Feasibility

All technologies used in the development of INFORM are open-source and free of licensing fees. The system was developed by the student research team as part of the academic capstone program, keeping the initial development costs to a minimum. Deployment costs are limited to basic server hosting, which is affordable and manageable within the school's existing information technology budget. The system eliminates the need for printed study load documents, reduces the time spent by school personnel responding to repetitive manual inquiries, and minimizes the risk of record loss or data duplication. In addition, the integrated JOBERT AI assistant helps reduce the volume of routine student inquiries by providing immediate responses to common concerns, thereby decreasing the workload of school

personnel. Therefore, the proposed system is considered economically feasible. thereby decreasing the workload of school personnel. Therefore, the proposed system is considered economically feasible.

Schedule Feasibility

The development of INFORM was completed in structured phases aligned with the academic capstone timeline, including requirements gathering, system design, development, integration and testing, and documentation. The system's modular architecture, with separate routes, controllers, and components for each module, allowed individual features to be developed and tested independently, ensuring that the project remained on schedule. Furthermore, the use of well-documented frameworks and development tools significantly reduced development time and simplified the debugging process. Therefore, the proposed system is considered feasible within the required project schedule.

Legal Feasibility

INFORM handles sensitive personal and academic data, including student names, grades, tuition records, and official documents. The system was developed in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines. Access to student records is strictly restricted based on user roles through JSON Web Token (JWT)-based authentication, ensuring that only authorized users can access or modify information within their assigned permissions. All user passwords are encrypted using bcrypt.js before being stored in the database, and no plain-text passwords are retained. In addition, every administrative action is recorded in an audit log table to ensure accountability and traceability. All third-party libraries

and frameworks used in the development of the system are released under permissive open-source licenses that permit academic and institutional use. Therefore, the proposed system is considered legally feasible.

Cost and Benefit Analysis